

Knowledge Retention & Catastrophic Forgetting of Large Language Models During Full Parameter Fine Tuning

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Introduction:

Large language models (LLMs) acquire vast amounts of factual and semantic knowledge during large-scale pretraining on web-scale corpora. This knowledge is encoded implicitly in the model weights, meaning that the pretrained representational space becomes the foundation for downstream performance. LLMs are often adapted to new objectives through different finetuning regimes. While these methods achieve strong improvements in task performance, they also introduce a well-documented challenge: catastrophic forgetting, where new training overwrites previously learned knowledge. Although this phenomenon is widely studied in smaller neural architectures, its mechanisms and severity in modern transformer-based LLMs remain poorly understood.

This project aims to empirically measure and characterize catastrophic forgetting in GPT-style transformer models during downstream finetuning. To do so, we implemented a full pipeline that (1) finetunes GPT-2 Small and GPT-2 Medium on the DailyDialog conversational dataset, (2) evaluates factual recall performance at multiple checkpoints throughout training using the T-REx subset of the LAMA benchmark, and (3) analyzes internal model behavior via layer-wise representation drift, computed using cosine similarity between hidden-state activations before and after finetuning. This combination of behavioral and internal metrics enables us to quantify how much knowledge is lost, how forgetting progresses over time, and which layers of the network are structurally rewritten during adaptation.

Our results show that while both GPT-2 Small and GPT-2 Medium exhibit measurable loss of factual recall, the impact differs substantially by scale: the finetuned GPT-2 Small model retrained only 56% of questions it correctly answered before finetuning, yet mostly maintained its pretrained representational structure, with only minor drift localized to the final layer. The GPT-2 Medium model experiences major restructuring across deeper layers, accompanied by substantial drops in factual accuracy, retaining only 61% of knowledge it used to answer correctly pre-finetuning. The findings suggest that catastrophic forgetting is not uniform, but rather occurs in stages depending on the statistical representation of the dataset and size of the model, and impacts deeper-layer structures more severely the longer the model is finetuned.

Background:

LLMs are built on the Transformer architecture, which uses self-attention mechanisms to compute contextual representations of text sequences. During pretraining, the model learns to predict the next token across billions of examples, implicitly internalizing statistical patterns about grammar, world knowledge, and reasoning. The Transformer stack consists of multiple layers, each transforming hidden representations to extract increasingly abstract semantic structure; lower layers typically encode lexical and syntactic information, while higher layers capture broader semantics and task-level decision patterns.

After pretraining, LLMs are typically adapted for downstream tasks using finetuning or more specialized methods such as instruction-tuning or reinforcement learning from human feedback (RLHF). Finetuning updates all model parameters on a narrower task-specific dataset (e.g., dialogue, summarization, or code

generation). While these methods enable strong specialization gains, they also risk catastrophic forgetting, in which finetuning overwrites or disrupts previously acquired information. Although catastrophic forgetting has been widely explored in broader neural network architectures, its behavior within large Transformer-based LLMs is less understood.

Recent studies (Kotha et al. 2023; Li et al. 2024; Luo et al. 2023; Meng et al. 2022) show that finetuning can degrade factual recall and that only a small subset of parameters may store factual associations. However, most of this work focuses on isolated fact-editing rather than full-model domain adaptation. To address this gap, we examine catastrophic forgetting during broad downstream finetuning by adapting GPT-2 models to the DailyDialog dataset and measuring changes in factual recall over training. Using both behavioral accuracy (by benchmarking across LAMA T-REx) and layer-wise representation drift, we analyze where forgetting occurs and how it progresses, with attention to differences across model scale.

Methods:

Experimental Setup:

We study catastrophic forgetting in two GPT-style transformer models: GPT-2 Small (117M parameters, 12 layers) and GPT-2 Medium (345M parameters, 24 layers). We first evaluate their baseline factual knowledge using the T-REx subset of the LAMA benchmark. Both models are then finetuned on an unrelated downstream objective (specifically dialogue generation using the DailyDialog dataset) for 3 and 6 training epochs, with checkpoints saved every 250 training steps. At each checkpoint, we re-evaluate factual recall on T-REx to measure forgetting over time. Additionally, we compute layer-wise representation drift, using cosine similarity between hidden-state activations on a set of 5,000 sampled T-REx prompts before and after finetuning, to analyze how internal representations evolve during adaptation.

Datasets:

T-Rex subset of LAMA	A knowledge-base completion benchmark containing subject-relation-object triples from Wikidata. For each triple, we generate cloze-style prompts (e.g., " <i>Kupreanof Island is located in the region of <mask></i> ") and evaluate whether the model predicts the correct object’s first token. After preprocessing and quality filtering, the resulting benchmark contained ~50,000 evaluated instances.
DailyDialog	A multi-turn conversational dataset designed for natural and emotional dialog modeling. It is unrelated to factual knowledge, making it an ideal candidate to isolate catastrophic forgetting. It contains ~13k dialogues and was used for full parameter finetuning.

Table 1: Benchmarking & Finetuning datasets

We also compute subject frequency statistics from Wikidata alias frequencies, which allow separating rare vs. common facts into buckets based on quartiles: *rare*, *medium*, *frequent*, and *very frequent*. This gives insight into what type of knowledge is more prone to being forgotten. Final evaluation for fact frequency robustness considers only *Rare* knowledge, and *Common* knowledge (the aggregate of *medium*, *frequent*, and *very frequent* facts).

Evaluation Metrics:

We compare each checkpoint against the pretrained baseline using four complementary metrics:

Metric	Meaning	Computation
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Top-k Accuracy	How often the correct token appears in the top-k predictions	Mean correctness across examples
Knowledge Retention Rate (KRR)	Fraction of facts originally answered correctly that remain correct after finetuning	$KRR = \frac{ S_0 \cap S_{ft} }{ S_0 }$ S_0 = the set of examples answered correctly at baseline (top-1) S_{ft} = the set of examples answered correctly after finetuning (top-1)
Fact Frequency Robustness (FFR)	Whether common or rare facts deteriorate differently	$FFR = KRR_{common} - KRR_{rare}$
Representation Drift	Change in internal representations compared to baseline	Cosine similarity of hidden states across transformer layers

Table 2: Metrics

Implementation Details:

The implementation of the project relied on numerous different libraries and frameworks; notably HuggingFace Transformers (model loading, tokenization, inference), HuggingFace Datasets (dataset storage and preprocessing), PyTorch (training & hidden state analysis), as well as NumPy & Pandas (metric computation). Finetuning was done under an NVIDIA RTX 5080 GPU. Finetuning kept batch size constant at 8, and learning-rate at 5e-5 with cosine decay. The optimizer used was AdamW.

Results:

Baseline Results:

Before applying any finetuning, the pretrained GPT-2 small and GPT-2 medium models achieved the following results on factual recall using the T-REx subset of the LAMA benchmark (~50K filtered examples):

Model	Top-1 Accuracy	Top-5 Accuracy	KRR	KRR-Rare	KRR-Common
GPT-2 Small	0.2697	0.5351	1.0	1.0	1.0
GPT-2 Medium	0.4942	0.6763	1.0	1.0	1.0

Table 3: Baseline Model Results

As expected, GPT-2 medium shows substantially higher factual recall at baseline; an 83.2% relative improvement in top-1 accuracy and ~27% improvement in top-5 accuracy compared to GPT-2 small. This confirms that model scale significantly increases the amount of parametric knowledge retained from pretraining. Both models demonstrate identical knowledge retention rates of 1.0 at this stage, as KRR is computed relative to baseline performance and therefore serves as an upper reference bound for comparison after training.

GPT-2 Small Post Finetuning:

After establishing baseline performance, we finetuned GPT-2 small on the DailyDialog dataset for approximately 2,000 training steps (3 epochs) and evaluated factual recall at 10 intermediate checkpoints. Figure 1 illustrates how the Knowledge Retention Rate (KRR) evolves over time relative to the pretrained baseline.

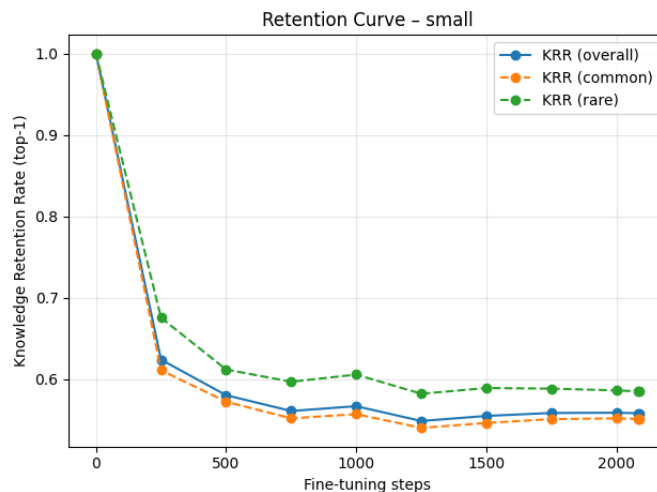


Figure 1: GPT-2 Small knowledge retention across finetuning

Finetuning leads to a rapid degradation in factual recall within the first 250 steps. KRR drops sharply from 1.0 at baseline to 0.624 after only 250 steps, representing an immediate ~38% loss of recoverable knowledge. Beyond this point, the degradation stabilizes, with KRR values fluctuating between 0.548 and 0.559 for the remainder of training.

Step	KRR	Accuracy (Top-1)	Accuracy (Top-5)
Baseline	1.0000	0.2697	0.5351
250	0.6239	0.2007	0.4419
500	0.5801	0.1882	0.4124
1250	0.5484	0.1984	0.4053
2086 (final)	0.5579	0.2044	0.4178

Table 4: Finetuned GPT-2 Small Results

This pattern demonstrates a form of catastrophic forgetting in which the majority of knowledge loss occurs early, followed by a long plateau where additional finetuning does not significantly worsen forgetting. We will confirm this later by analyzing representation drift against KRR.

One surprising result shown in the retention curve is that ‘rare’ facts (i.e. knowledge that occurs in small portions of the training data) does not degrade faster than common facts. Indeed, this can be confirmed with the FFR curve:

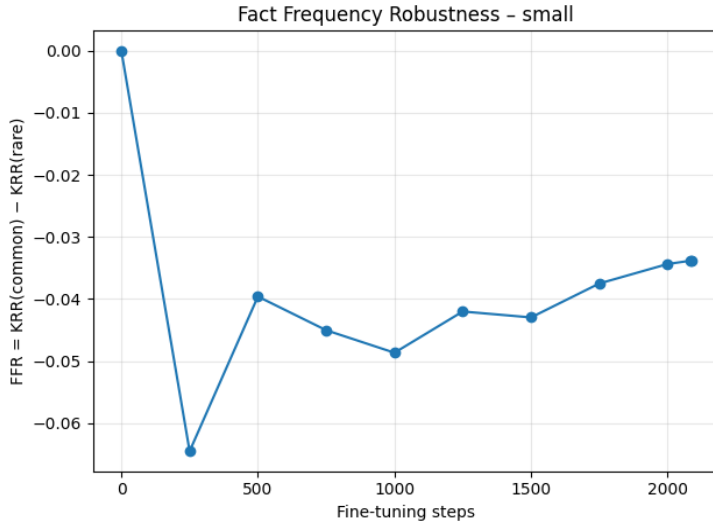


Figure 2: GPT-2 Small FFR across finetuning

Step	KRR-Rare	KRR-Common	FFR
Baseline	1.000	1.000	0.000
250	0.675	0.611	-0.0645
1000	0.606	0.557	-0.0487
Final	0.585	0.551	-0.0338

Table 5: GPT-2 Small FFR across finetuning

These negative values indicate that rare facts are initially more resilient than common ones, contradicting intuitive assumptions that rare knowledge should be erased first. One interpretation is that common knowledge participates more frequently in training contexts and is therefore more exposed to being overwritten, while rare facts reside in narrower parameter subspaces that finetuning does not initially disrupt. However, we will see later that this trend isn't universal when training over more epochs, in the case of GPT-2 medium.

The layer-wise hidden state cosine similarity between the baseline model and each finetuned checkpoint also yielded valuable results. Figures 3 shows that drift primarily occurs in the final few transformer layers, while lower layers remain structurally stable.

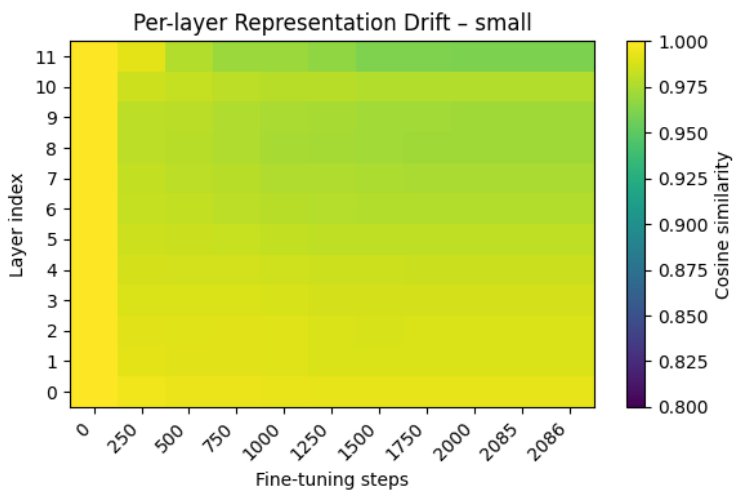


Figure 3: GPT-2 small per-layer representation drift

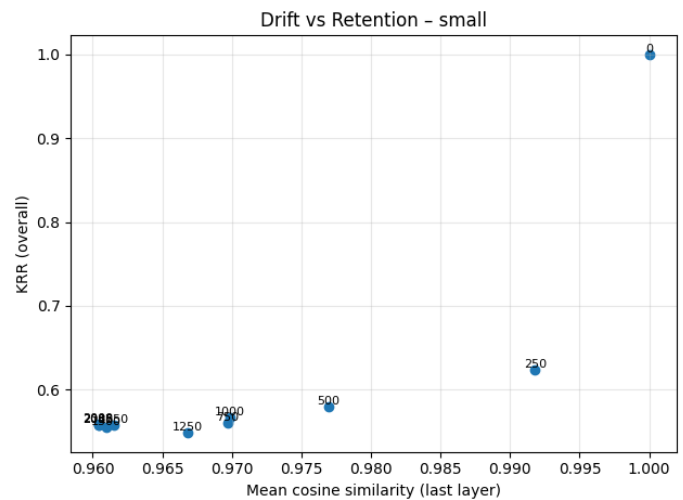


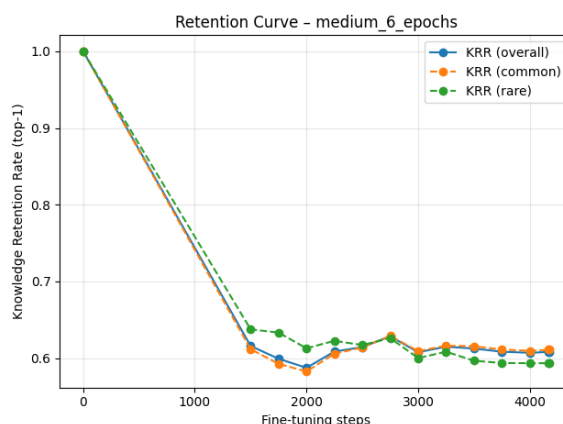
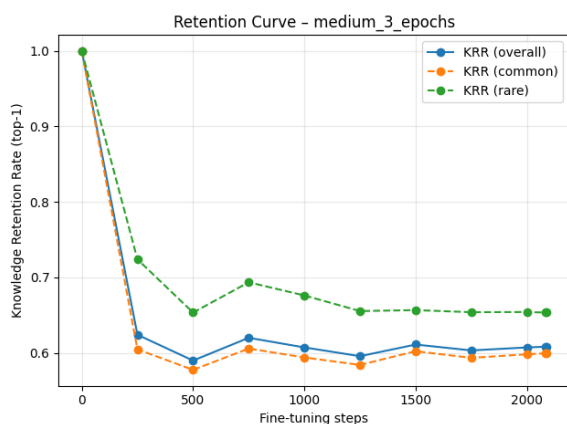
Figure 4: GPT-2 small drift vs retention

The cosine similarity of the last layer drops from 1.0 at baseline to ~0.96 after finetuning, while earlier layers remain above 0.98. My interpretation is that finetuning rewrites higher-level semantic representations while preserving lower-level lexical and syntactic structure. Additionally, Figure 4

confirms that forgetting plateaus; in the last layer, the internal representation drift from the baseline seems to converge at some point. KRR decays at an exponential rate during the finetuning process. Forgetting happens quickly and drastically early during finetuning.

GPT-2 Medium Post Finetuning:

After finetuning for 3 epochs, knowledge retention deteriorated significantly. Overall KRR dropped from 1.0 to 0.608, representing a 39% reduction in originally correct factual responses. Extending training to 6 epochs did not substantially worsen forgetting: final KRR remained stable around ~0.608, and Top-1 accuracy plateaued at 33.9%, similar to the 3-epoch model. Like with the GPT-2 small model, rare-fact retention was initially better preserved than common-fact retention. This indicates that common facts are overwritten more aggressively during early finetuning, while rarer facts initially remain more stable. One thing that is interesting is that when finetuned across more epochs, eventually the KRR associated with rare facts decreases at a more aggressive rate.



Figures 5 & 6: GPT-2 Medium knowledge retention across finetuning regimes

This leads to an interesting empirical insight. Finetuning does not cause uniform forgetting; it unfolds in stages. Initial adaptation overwrites broad common knowledge, while continued optimization dismantles more specialized and rare factual structures that are hidden deeper within the network.

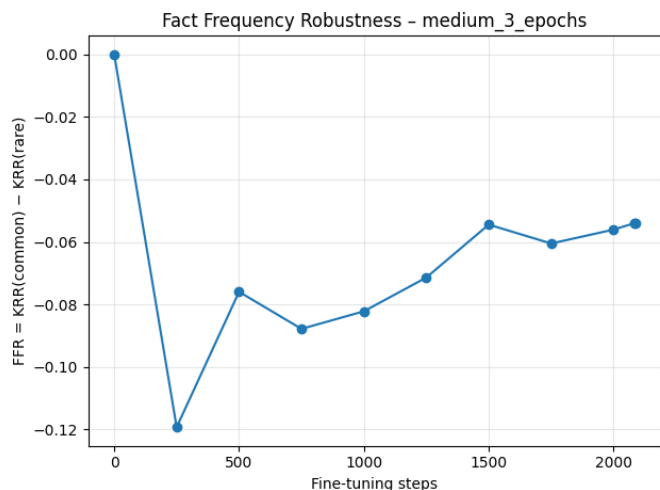


Figure 7: FFR for GPT-2 Medium under 3 epoch finetuning regime

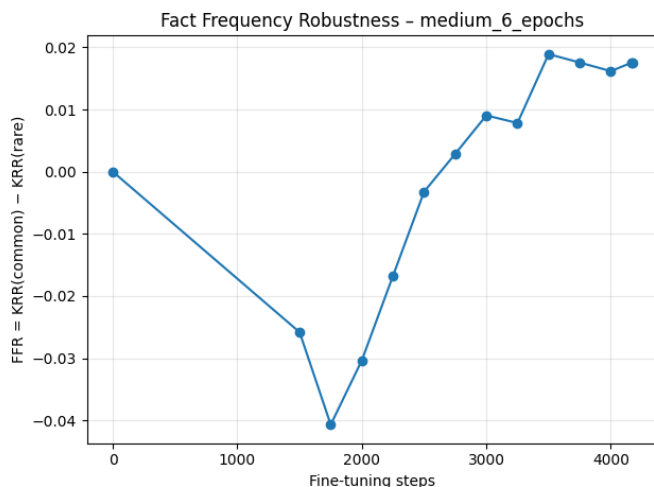


Figure 8: FFR for GPT-2 Medium under 6 epoch finetuning regime

The change in the model's internal representations during finetuning also highlight an interesting observation; notably it highlights where the forgetting occurs within the architecture. The per-layer drift heatmaps show that for GPT-2 Medium, the last layer undergoes extremely large representational shifts, while early and middle layers remain largely unchanged.

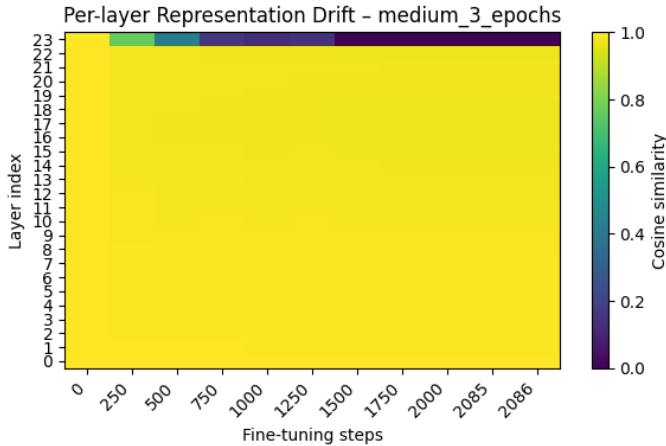


Figure 9: GPT-2 Medium per-layer representation drift (3 epochs)

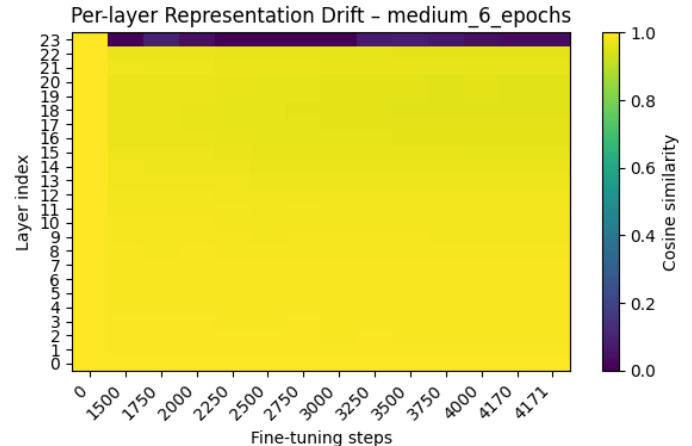


Figure 10: GPT-2 Medium per-layer representation drift (6 epochs)

The heatmaps reveal that drift is highly localized to the final layers. Layers 0-22 remain >0.95 similarity, while layer 23 collapses nearly to zero, indicating catastrophic representational restructuring near output decoding. This supports the hypothesis that finetuning primarily re-specializes the readout layers, leaving earlier linguistic feature layers intact. The extent of the difference in cosine similarity between the baseline and finetuned models in the last layer is shown by figures 11 & 12:

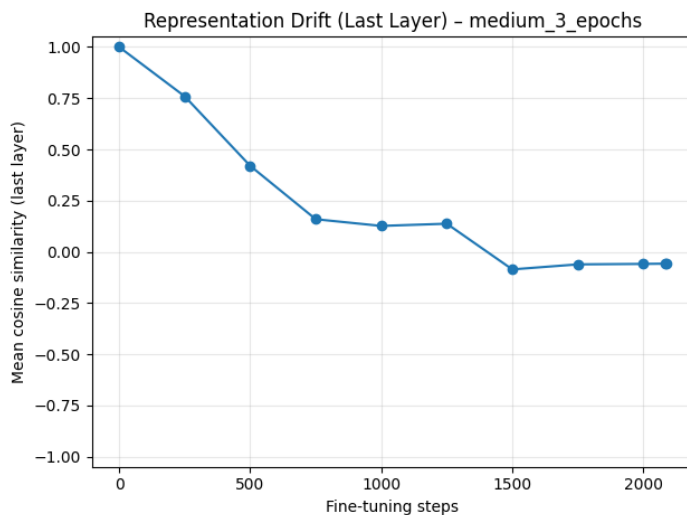


Figure 11: GPT-2 Medium last layer representation drift (3 epochs)

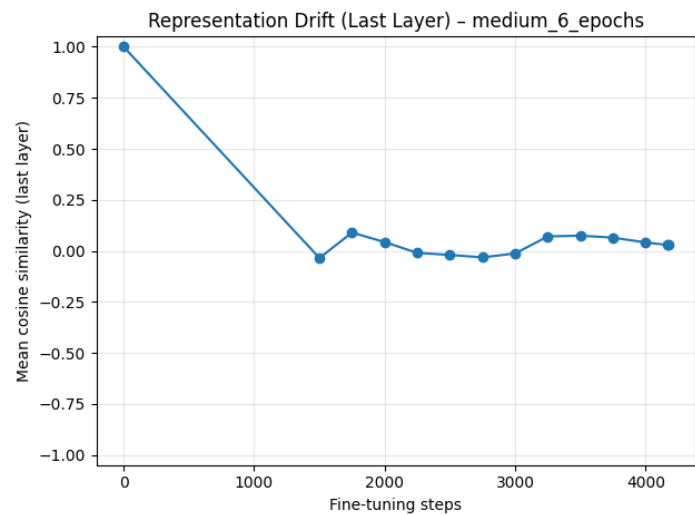


Figure 12: GPT-2 Medium last layer representation drift (6 epochs)

As we can see, the cosine similarity of the finetuned GPT-2 Mediums is very different from the baseline model. In fact, we see that for the 3 epoch model the cosine similarity crossed below zero, meaning that the new representation is almost orthogonal with the original space.

And finally, the exponential decay trajectory we see with KRR throughout the finetuning process is confirmed in the GPT-2 Medium finetuned models as well:

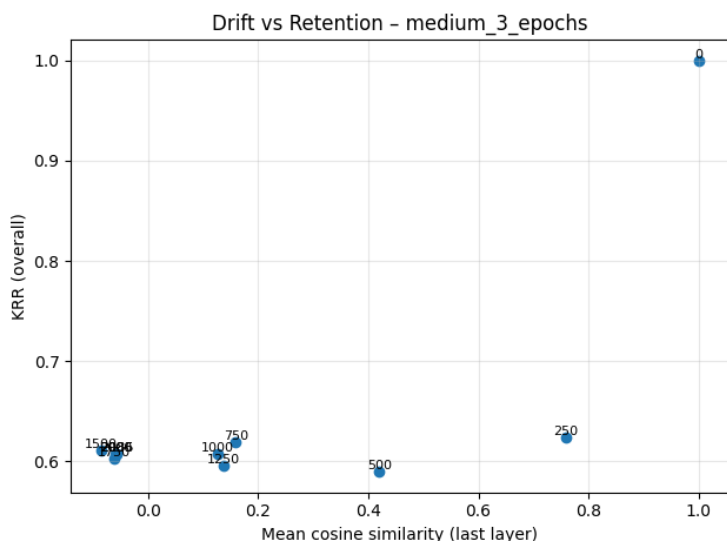


Figure 13: GPT-2 Medium drift vs retention (3 epochs)

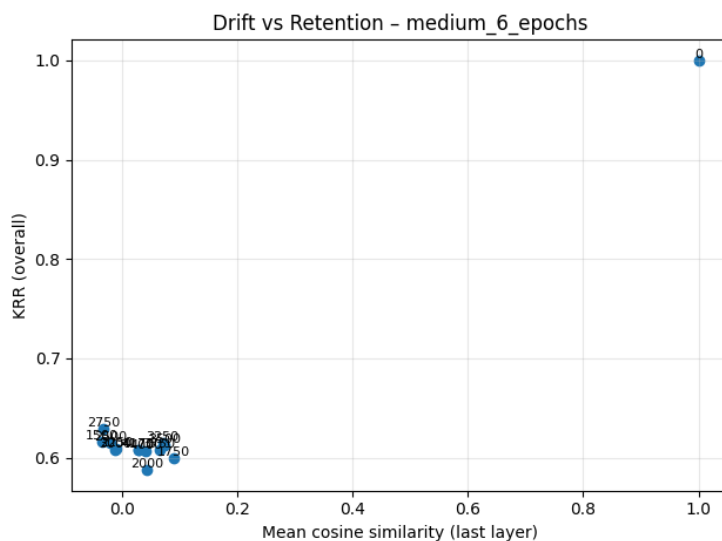


Figure 14: GPT-2 Medium drift vs retention (3 epochs, only values after the 3-epoch mark shown)

Conclusions:

Using the T-REx factual QA benchmark, we found that finetuning causes measurable degradation in factual recall for both models. GPT-2 Small retained 56% of its originally correct facts (final KRR ≈ 0.56) and exhibited only mild representational drift localized to the final transformer layer. GPT-2 Medium retained more knowledge overall (final KRR ≈ 0.61), yet experienced dramatically larger representational changes: the last layer's cosine similarity dropped near zero (and below zero in the 3-epoch run), indicating near-complete reconfiguration of the output-decoding space. Despite this dramatic internal shift, core earlier layers remained highly stable (>0.95 similarity), suggesting that finetuning rewrites readout layers while preserving pretrained semantic structure. These findings suggest that catastrophic forgetting in LLMs is not uniform across layers or fact types. Forgetting occurs rapidly early in training and then stabilizes, and common facts are overwritten before rare ones, with rare-fact degradation emerging only after more extensive finetuning. One interpretation of this result is that the model overwrites broad, high-frequency knowledge first because these representations are densely encoded and strongly activated by the training objective. Rare facts are initially more stable because they occupy more isolated or sparsely used regions of the representation space. However, as training continues and the model reallocates internal capacity to specialize for the new task, this sparsity advantage erodes: the network must reorganize weights more aggressively, increasing the probability that these sparsely encoded memories are overwritten. Overall, larger models appear more resilient to forgetting, even while undergoing greater representational drift.

Impact and Future Work:

I believe the results have practical implications for model training workflows, and indicate that preserving pretrained knowledge may require protecting late-stage representations rather than earlier ones. For example, if finetuning overwrites guardrails or ethical reasoning embedded in the base model, harmful behaviors could re-emerge unintentionally. Knowledge of what gets forgotten and where could help in stabilizing public-facing model consistency.

Future work could explore parameter-efficient finetuning techniques to determine whether isolating parameter updates can reduce representational drift and improve knowledge retention. Given more time, I think this type of analysis could lead to the formulation of some kind of ‘empirical forgetting’ scaling law; as we finetune across more data and with more epochs, where can we expect the greatest shift in the model’s internal representation to be? Obviously, this is a tremendously difficult task, but I believe it potentially is doable given certain assumptions.

References:

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